

Your grade: 96.66%

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Next item →

1.

Which algorithm does Keras Tuner NOT use for hyperparameter optimization?

1 / 1 point
- ☐

Random Search

☐

Bayesian Optimization

☐

Hyperband

☒

Genetic Algorithms
-  Correct

Correct! Keras Tuner does not use genetic algorithms for hyperparameter optimization.

2.

Select all the steps involved in hyperparameter tuning using the Keras Tuner library.

0.8 / 1 point

☐

Train the model on the entire dataset without validation.

☒

Run the search to find the best hyperparameters.

 Correct

Correct! Running the search is necessary to identify the optimal hyperparameters.

☒

Specify the hyperparameters to be tuned.

 Correct

Correct! Specifying hyperparameters is essential for tuning.

☒

Randomly initialize all weights and biases of the model.

 This should not be selected

Incorrect. While weights and biases initialization is necessary, it is not a step specific to Keras Tuner's hyperparameter tuning.

☒

Define the model structure.

 Correct

Correct! Defining the model structure is a crucial step in hyperparameter tuning.

☒

Choose the search algorithm (e.g., random search).

 Correct

Correct! Selecting an appropriate search algorithm is a key step in the process.

3.

Which of the following are benefits of using transfer learning?

1 / 1 point

☒

Reduced computational cost.

 Correct

Correct! Transfer learning can reduce computational costs by leveraging pre-trained models.

☐

Elimination of the need for data preprocessing.

☐

Ability to solve problems in completely unrelated domains.

☒

Faster convergence during training.

 Correct

Right! Using a pre-trained model can help the new model converge faster.

☒

Improved model accuracy with less data.

 Correct

Good job! Transfer learning can lead to better accuracy even with less training data.

4.

Which scenarios are appropriate for using pretrained deep learning models?

1 / 1 point

☐

Training a model from scratch with a massive dataset

☒

Transfer learning for a related task

 Correct

Correct! Transfer learning is a common use case for pretrained models.

☒

Predicting image classes with limited data

 Correct

Correct! Pretrained models can perform well even with limited data.

☐

Replacing traditional machine learning algorithms

☒

Feature extraction for a new task

 Correct

Well done! Pretrained models are excellent for feature extraction in new tasks.

5.

What does VGG stand for in the context of deep learning models?

1 / 1 point

☐

Variable Grid Generator

☐

Vector Gradient Group

☐

Visual Gradient Generator

☒

Visual Geometry Group

 Correct

Correct! VGG stands for Visual Geometry Group, the group that developed these models.